

ACCESSORY ECOSYSTEM, DOCKING SYSTEMS, AND EXPANSION HARDWARE EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device may operate as part of an expandable accessory ecosystem comprising supporting hardware, calibration fixtures, storage systems, charging solutions, and auxiliary sensing modules. The accessory ecosystem enables functional expansion of the device without modification of the core sensing architecture.

Accessory embodiments may include docking stations configured for charging, data synchronization, storage, calibration verification, or automated maintenance routines. Docking systems may provide electrical charging contacts, inductive charging interfaces, wireless communication synchronization, or environmental conditioning such as drying or sterilization assistance.

In certain embodiments, docking stations may incorporate calibration references allowing automatic verification of distance sensing accuracy or compliance sensing response when the device is placed within the dock. Automated calibration checks performed during docking may improve long-term measurement consistency and reduce user burden.

The accessory ecosystem may include removable handles, interchangeable insertable modules, or attachable sensing extensions configured to modify device capability. Modular expansion hardware may allow upgrade paths ranging from simplified consumer configurations to enhanced research or clinical-support implementations.

Storage accessories may include protective cases, hygienic containment systems, travel enclosures, or smart storage devices capable of monitoring device status, battery level, cleaning compliance, or usage intervals. Smart storage systems may communicate with associated software to provide reminders or maintenance notifications.

In certain embodiments, accessory hardware may support automated inflation preparation, pressure equalization, or sensor warm-up routines prior to measurement sessions. Preparation accessories may therefore standardize operating conditions before data acquisition begins.

The ecosystem may further include training or demonstration accessories designed for onboarding, simulation, or educational purposes. Such accessories may replicate measurement workflows without collecting physiologic data, enabling safe user familiarization.

Additional accessory embodiments may incorporate auxiliary sensors positioned externally to the primary device. External sensing modules may capture environmental context, body position information, or complementary physiologic signals synchronized with intraluminal measurements.

In certain implementations, accessory components may communicate through wired connections, short-range wireless protocols, or secure device pairing systems. Accessories may therefore operate as coordinated nodes contributing to a broader measurement platform.

Accordingly, the disclosed accessory ecosystem enables scalable expansion of device functionality, supports long-term usability, simplifies maintenance, and establishes a modular hardware framework capable of evolving alongside future sensing technologies and computational capabilities.